

Topic 9

Statistics

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

9.1	Classifying and interpreting data
9.2	Averages and range
9.3	Frequency tables and estimated mean
9.4	Cumulative frequency and box plots
9.5	Histograms and frequency density
9.6	Scatter diagrams and correlation

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Kinds of data and displays

Categorical data name groups; numerical data are discrete counts or continuous measurements. Choose a bar chart for categories, a histogram for grouped continuous values and a scatter diagram for paired measurements. An unequal interval width in a histogram requires frequency density, not bar height equal to frequency.

Worked example

A survey records number of siblings and height. Sibling count is discrete; height is continuous.

Exam check: Read the axis scale before inferring a trend or an unusual value.

2. Mean, median and spread

Mean = sum of values/number of values. Median is the middle ordered observation; mode is the most frequent. Range = largest - smallest. Quartiles split ordered data, and interquartile range = $Q3 - Q1$. Compare typical value and spread together.

Worked example

Data 2,3,3,6,11: sum = 25, mean = 5, median = 3, range = 9.

Exam check: Do not choose the median before ordering raw data.

3. Grouped data and estimated means

For a frequency table, multiply each exact value by frequency before dividing by total frequency. For grouped intervals, replace each interval with its midpoint; the resulting mean is an estimate. The modal class is the interval with highest frequency, or highest density when widths differ.

Worked example

Groups 0-10 ($f=2$) and 10-20 ($f=3$) have midpoints 5 and 15. Estimated mean = $(2 \times 5 + 3 \times 15)/5 = 11$.

Exam check: A grouped mean is generally approximate because individual observations are unknown.

4. Cumulative frequency and box plots

Cumulative frequency adds frequencies up to an upper class boundary. For n observations, estimate $Q1$ at $n/4$, median at $n/2$ and $Q3$ at $3n/4$ on a cumulative frequency graph. A box plot shows minimum, lower quartile, median, upper quartile and maximum.

Worked example

For $Q1 = 12$ and $Q3 = 28$, the interquartile range is 16.

Exam check: The width of the box is the IQR; the full whisker-to-whisker span is the range.

5. Histograms and correlation

Frequency density = frequency/class width. The rectangle area represents frequency, so compare area, not just height, when widths differ. A scatter diagram may show positive, negative or no correlation. A line of best fit estimates within the data range; extrapolation outside can be unreliable.

Worked example

For a class width 5 with frequency 20, density = 4. A width 10 class with frequency 20 has density 2.

Exam check: Correlation alone does not prove causation.

6. Sampling and interpretation

A random sample gives each member a known chance of selection; a convenience sample may be biased. Large samples reduce random fluctuation but do not automatically fix selection bias. Interpret what a graph does and does not support, and include units for estimates.

Worked example

Surveying only early arrivals about school transport excludes some late commuters and may bias the result.

Exam check: A precise calculation from a biased sample may still misrepresent the population.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 9 · Statistics

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 28 marks

Section A · Multiple choice

Choose one answer for each question.

1. Median of 2,4,9? [1]

- A 2
- B 4
- C 5
- D 9

2. Mean of 3,5,7,9? [1]

- A 5
- B 6
- C 7
- D 8

3. Range of 2,5,11,14? [1]

- A 9
- B 11
- C 12
- D 14

4. For grouped data, the mean uses [1]

- A lower bounds only
- B class midpoints
- C upper bounds only
- D modal class only

5. Frequency density equals [1]

- A width/frequency
- B frequency $\times$ width
- C frequency/width
- D frequency + width

6. A box plot box spans [1]

- A minimum to maximum
- B Q1 to Q3
- C Q1 to median
- D median to maximum

7. A downward scatter trend is [1]

- A positive correlation
- B negative correlation
- C no correlation
- D causation

8. Which is a discrete numerical variable? [1]

- A height
- B mass
- C number of books
- D temperature

Section B · Structured questions

9. Five test scores are 3, 4, 4, 7 and 12.

(a) Find the mean. [2]

(b) Find the median and mode. [2]

(c) Find the range. [1]

[Total: 5]

10. Grouped heights have intervals 140-150 with frequency 3, 150-160 with frequency 5, and 160-170 with frequency 2.

(a) Write the three class midpoints. [2]

(b) Estimate the mean height. [3]

(c) State why the answer is an estimate.

[1]

[Total: 6]

11. A distribution has minimum 2, $Q1 = 8$, median 14, $Q3 = 23$, maximum 31.

(a) Find the interquartile range.

[1]

(b) Find the range.

[1]

(c) State the five values used in a box plot in order.

[2]

[Total: 4]

12. Class 0 ■ $x < 5$ has frequency 15 and class 5 ■ $x < 15$ has frequency 20.

(a) Calculate each frequency density.

[2]

(b) State the relative heights of the two histogram bars.

[1]

(c) Explain why their areas represent frequencies of 15 and 20.

[2]

[Total: 5]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. B — 4. The middle of the ordered list is 4.
2. B — 6. $(3+5+7+9)/4 = 6$.
3. C — 12. Largest minus smallest is $14 - 2 = 12$.
4. B — class midpoints. Midpoints approximate the unknown values within each interval.
5. C — frequency/width. The histogram height equals frequency divided by class width.
6. B — Q1 to Q3. The box begins at the lower quartile and ends at the upper quartile.
7. B — negative correlation. As one variable increases, the other tends to decrease.
8. C — number of books. The number of books is a count of whole items.

Section B · Worked solutions

9. a) $(3+4+4+7+12)/5 = 6$ [2]. b) Ordered data give median 4 and mode 4 [2]. c) $12 - 3 = 9$ [1].

10. a) Midpoints are 145, 155, 165 [2]. b) $(3 \times 145 + 5 \times 155 + 2 \times 165)/10 = (435 + 775 + 330)/10 = 154$ cm [3].
c) Actual observations within each class are unknown; using midpoints approximates them [1].

11. a) $23 - 8 = 15$ [1]. b) $31 - 2 = 29$ [1]. c) 2, 8, 14, 23, 31 [2].

12. a) $15/5 = 3$, $20/10 = 2$ [2]. b) First bar height 3, second bar height 2 [1]. c) Frequency = density $\times$ width, giving $3 \times 5 = 15$ and $2 \times 10 = 20$ [2].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and
examiner report provided by the user. All practice questions are original.